

JEE MAIN REVISION MASTER: SURFACE CHEMISTRY OUT OF SYLLABUS

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 113 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Surface Chemistry — Core Principles & Governing Laws	75 PYQs	Core Principle
Very High	Surface Chemistry — Applied Cases, Reactions & Traps	21 PYQs	Applied Topics
Very High	Surface Chemistry — Standard Calculations & Problem Archetypes	17 PYQs	Quantitative Analysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
Surface Chemistry — Core Principles & Governing Laws	$w_1, \quad \frac{x}{m} = kP^{1/n}, \quad \frac{1}{n}$
Surface Chemistry — Applied Cases, Reactions & Traps	$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g), \quad 4NH_3(g) + 5O_2(g) \rightarrow 4NO(g) + 6H_2O(g), \quad N_2(g) + 3H_2(g) \xrightarrow{Fe(s)} 2NH_3(g)$
Surface Chemistry — Standard Calculations & Problem Archetypes	$\approx 17 \times 10^{-2}, \quad AlCl_3, \quad 41.4 kJ mol^{-1}$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Surface Chemistry — Core Principles & Governing Laws (Very High Yield — 75 PYQs)

Core Principle: Fundamental definitions, core postulates, and governing physical/chemical relations in Surface Chemistry.

■ **Caution / JEE Pitfall:** Carefully verify units, standard reference states, and sign conventions in Surface Chemistry.

■ **Speed Trick:** Focus on high-yield formulas and standard limiting approximations tested in Surface Chemistry PYQs.

Representative Source PYQs:

- [2024 Evening Shift] Q#1 (MCQ): "The adsorbent used in adsorption chromatography is/are - A. silica gel B. alumina C. quick lime D. magnesia Choose the most appropriate answ..." [Ans: N/A]
- [2024 Morning Shift] Q#2 (MCQ): "'Adsorption' principle is used for which of the following purification method?..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : In expensive scie..." [Ans: N/A]

3.2 Surface Chemistry — Applied Cases, Reactions & Traps (Very High Yield — 21 PYQs)

Core Principle: Special cases, experimental observations, and conceptual traps designed to test deep understanding of Surface Chemistry.

■ **Caution / JEE Pitfall:** Read questions carefully for keywords like 'incorrect', 'not true', or non-standard conditions.

■ **Speed Trick:** Elimination of extreme or unphysical answer choices improves accuracy in assertion-reason and multiple-statement questions.

Representative Source PYQs:

- [2023 Morning Shift] Q#1 (MCQ): "Four gases A, B, C and D have critical temperatures 5.3, 33.2, 126.0 and 154.3K respectively. For their adsorption on a fixed amount of ch..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (Numerical): "Based on the given figure, the number of correct statement/s is/are _____. A. Surface tension is the outcome of equal attractive and re..." [Ans: 2]
- [2023 Evening Shift] Q#2 (MCQ): "Ferric chloride is applied to stop bleeding because -..." [Ans: D]

3.3 Surface Chemistry — Standard Calculations & Problem Archetypes (Very High Yield — 17 PYQs)

Core Principle: Quantitative numerical problem-solving techniques and standard question patterns frequently seen in Surface Chemistry.

■ **Caution / JEE Pitfall:** Watch for algebraic rounding, stoichiometric balancing, and boundary condition checks.

■ **Speed Trick:** Dimensional consistency checks and ratio comparisons can quickly eliminate incorrect options.

Representative Source PYQs:

- [2023 Evening Shift] Q#1 (MCQ): "In figure, a straight line is given for Freundlich Adsorption ($y=3x+2.505$). The value of $(1/n)$ and $\log K$ are respectively..." [Ans: N/A]
- [2023 Morning Shift] Q#1 (MCQ): "Statement I : For colloidal particles, the values of colligative properties are of small order as compared to values shown by true solutions..." [Ans: N/A]
- [2023 Evening Shift] Q#1 (Numerical): "Coagulating value of the electrolytes $AlCl_3$ and $NaCl$ for As_2S_3 are 0.09 and 50.04 respectively. The coagulating power of $AlCl_3$ is..." [Ans: 556]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2023 (Online) 12th April Morning Shift (Q#1)

Four gases A, B, C and D have critical temperatures 5.3, 33.2, 126.0 and 154.3 K respectively. For their adsorption on a fixed amount of charcoal, the correct order is:

- (A) $A > C > B > D$
- (B) $B > D > A > C$
- (C) $C > A > B > D$
- (D) $D > B > A > C$

Correct Answer: None

Solution Analysis: The extent of adsorption of a gas on a solid surface depends on the critical temperature of the gas and the temperature of the solid surface. The critical temperature is the temperature above which a gas cannot be liquefied, no matter how much pressure is applied. Therefore, gases with higher critical temperatures are less likely to be adsorbed on a solid surface than gases with lower critical temperatures. In this case, the gases are A, B, C, and D, with critical temperatures of 5.3 K, 33.2 K, 126.0 K, and 154.3 K, respectively. Therefore, the correct order of adsorption on a fixed amount of

Benchmark Problem 2: JEE Main 2023 (Online) 1st February Evening Shift (Q#1)

In figure, a straight line is given for Freundlich Adsorption ($y = 3x + 2.505$). The value of $(1/n)$ and $\log K$ are respectively.

- (A) 0.3 and $\log 2.505$
- (B) 3 and 2.505
- (C) 3 and 0.7033
- (D) 0.3 and 0.7033

Correct Answer: None

Solution Analysis: $(x/m) = kP^{1/n}$ $\log(x/m) = \log k + (1/n)\log P$ On comparing with $y = 3x + 2.505$, we get $(1/n) = 3$ and $\log k = 2.505$

Benchmark Problem 3: JEE Main 2023 (Online) 30th January Morning Shift (Q#1)

Given below are two statements : one is labelled as Assertion (A) and the other is labelled as Reason (R). Assertion (A) : In expensive scientific instruments, silica gel is kept in watch-glasses or in semipermeable membrane bags. Reason (R) : Silica gel adsorbs moisture from air via adsorption, thus protects the instrument from water corrosion (rusting) and / or prevents malfunctioning. In the light of the above statements, choose the correct answer from the options given below :

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A)
- (B) (A) is true but (R) is false
- (C) (A) is false but (R) is true
- (D) Both (A) and (R) are true but (R) is not the correct explanation of (A)

Correct Answer: None

Solution Analysis: Assertion is correct and Reason is correct explanation of Assertion. Silica gel adsorbs moisture and thus protects the instrument from water corrosion (rusting) and prevents malfunctioning.